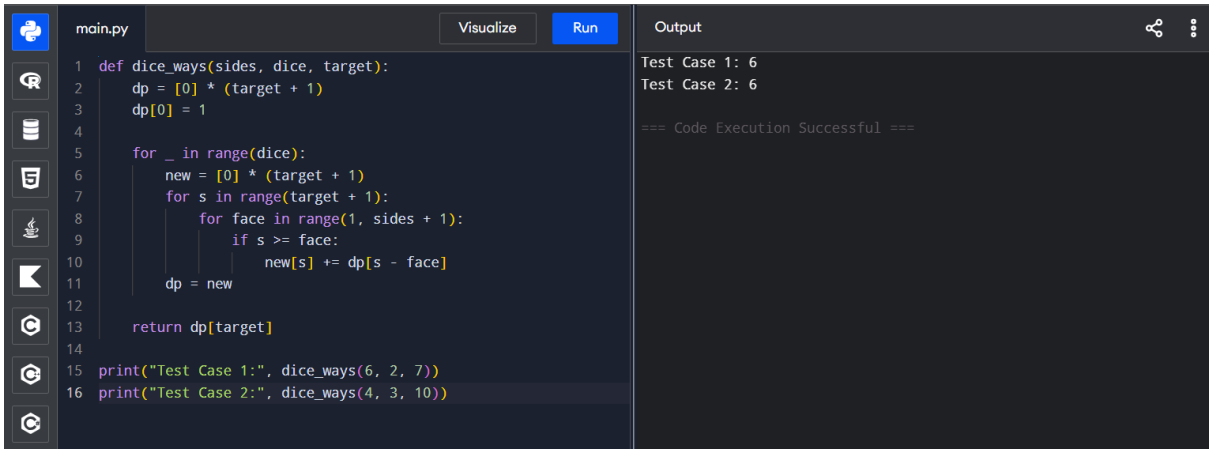
CHAPTER – 4

Name: Jeslet Jessica. T

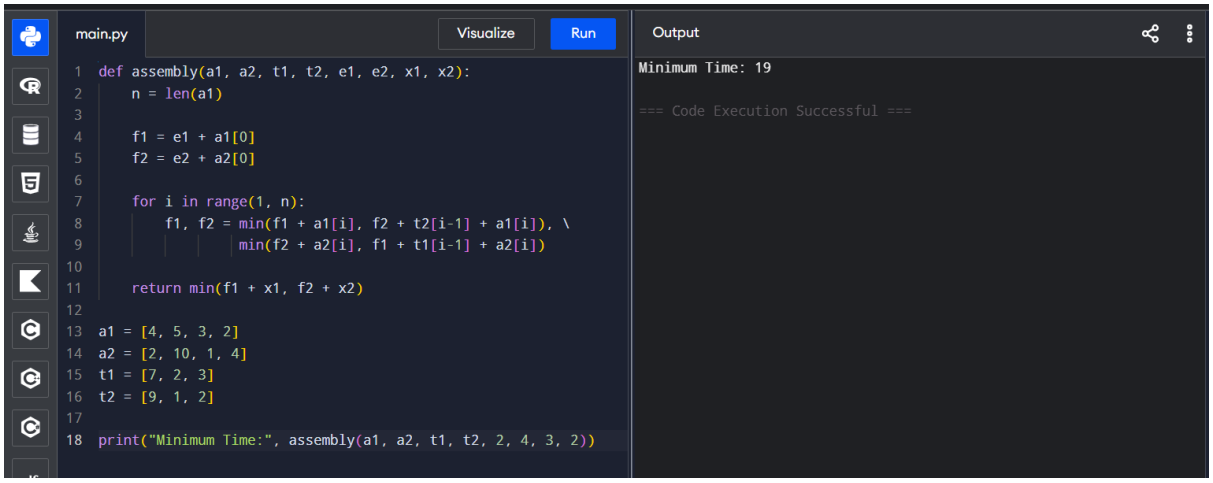
Regno: 192524135

Course: CSA0603– DAA

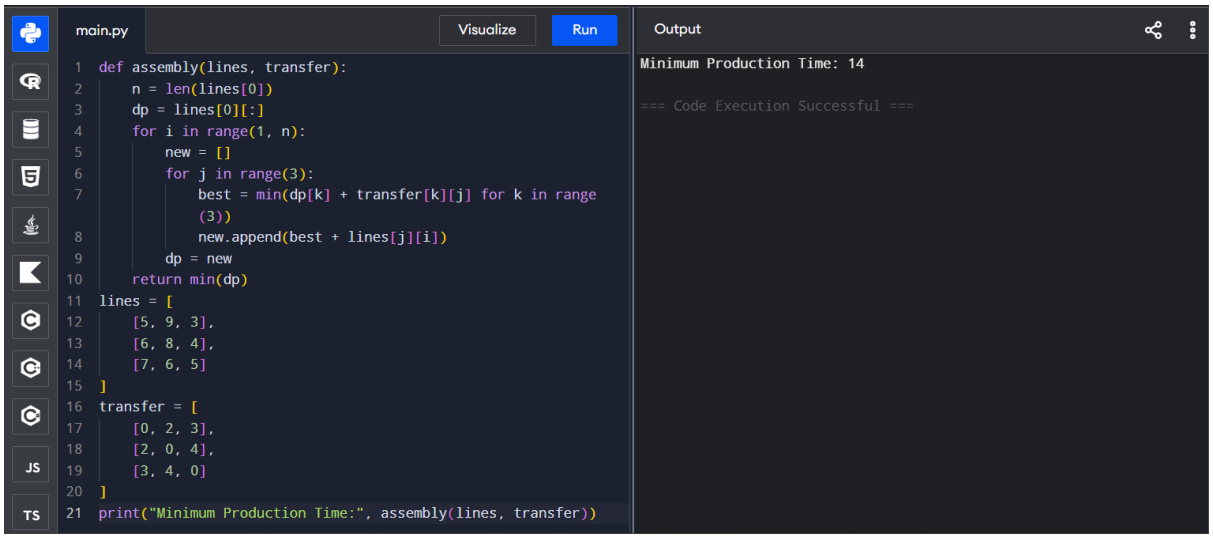
EXPERIMENT – 1



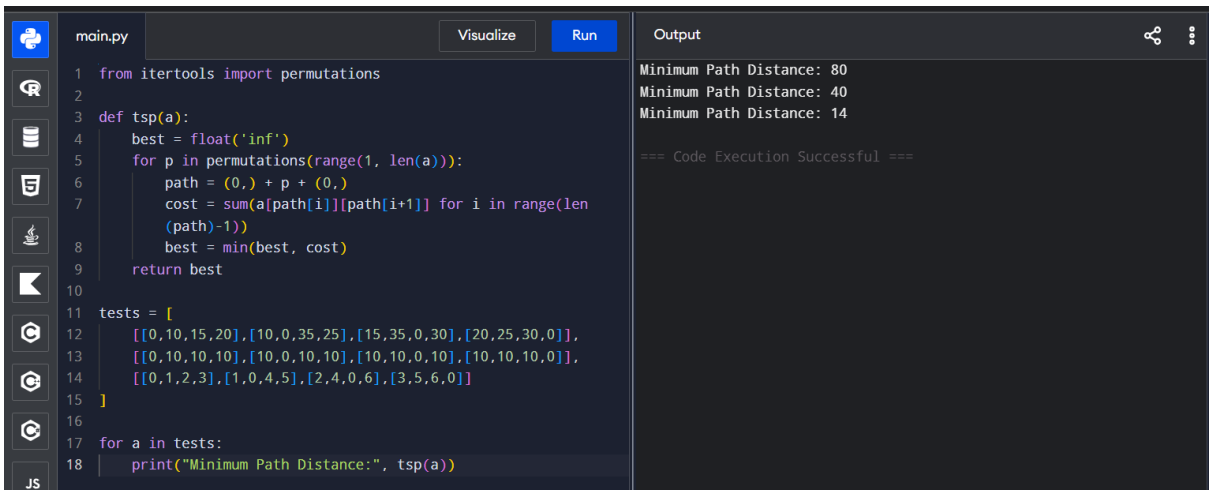
EXPERIMENT – 2



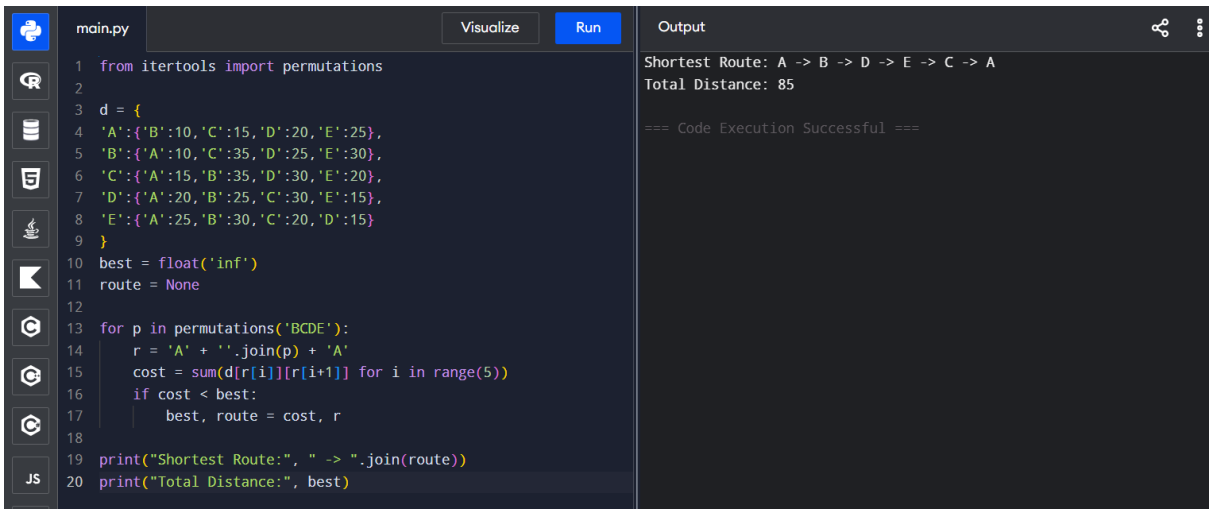
EXPERIMENT – 3



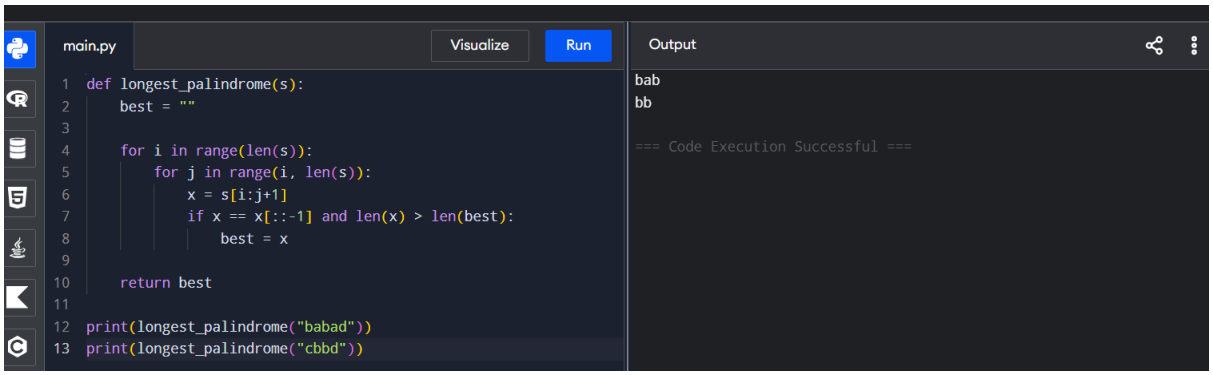
EXPERIMENT – 4



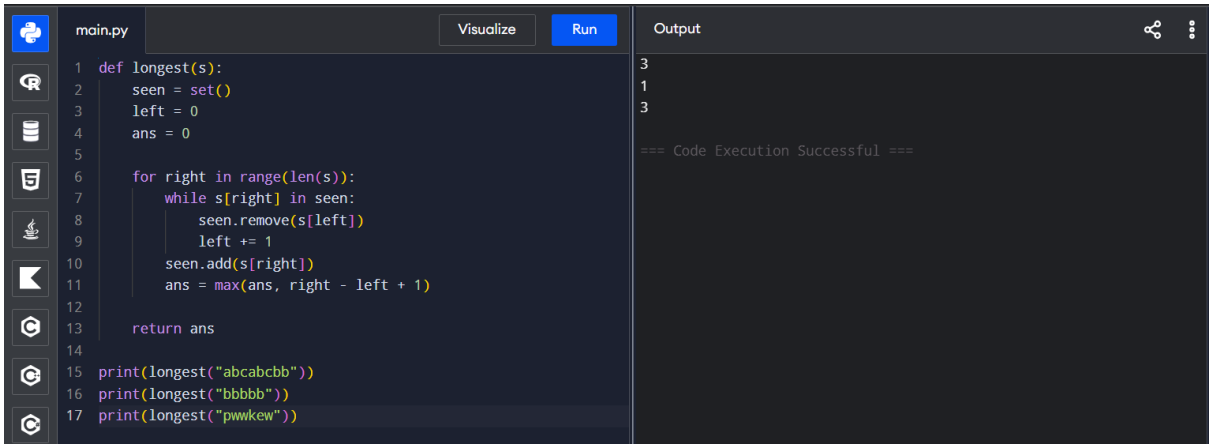
EXPERIMENT – 5



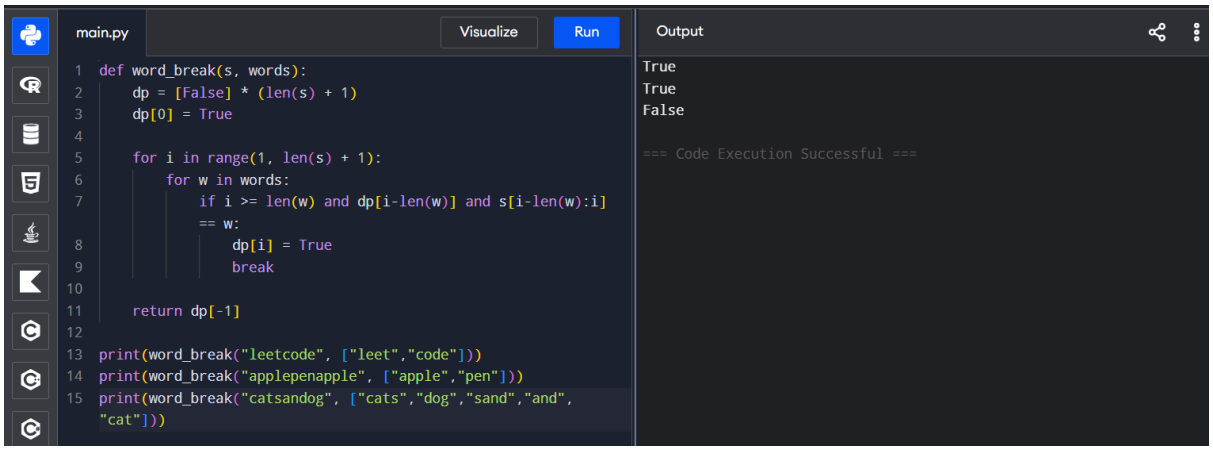
EXPERIMENT – 6



EXPERIMENT – 7



EXPERIMENT – 8



EXPERIMENT – 9

main.py

```python
def word_break(s, words):
    dp = [False] * (len(s) + 1)
    dp[0] = True

    for i in range(1, len(s) + 1):
        for w in words:
            if s[i-len(w):i] == w and dp[i-len(w)]:
                dp[i] = True
                break

    return dp[-1]

words = ["i","like","sam","sung","samsung","mobile",
         "ice","cream","icecream","man","go","mango"]

print("Yes" if word_break("ilike", words) else "No")
print("Yes" if word_break("ilikesamsung", words) else "No")
```

Output

```
Yes
Yes

=== Code Execution Successful ===
```

EXPERIMENT – 10

main.py

```python
def justify(words, width):
    res, line, length = [], [], 0
    for word in words:
        if length + len(word) + len(line) > width:
            spaces = width - length
            for i in range(spaces):
                line[i % (len(line)-1)] += " "
            res.append("".join(line))
            line, length = [], 0
        line.append(word)
        length += len(word)
    last = " ".join(line)
    res.append(last + " " * (width - len(last)))
    return res
words = ["This","is","an","example","of","text",
"justification."]
for x in justify(words, 16):
    print(x)
```

Output

```
This    is    an
example  of text
justification.

=== Code Execution Successful ===
```

EXPERIMENT – 11

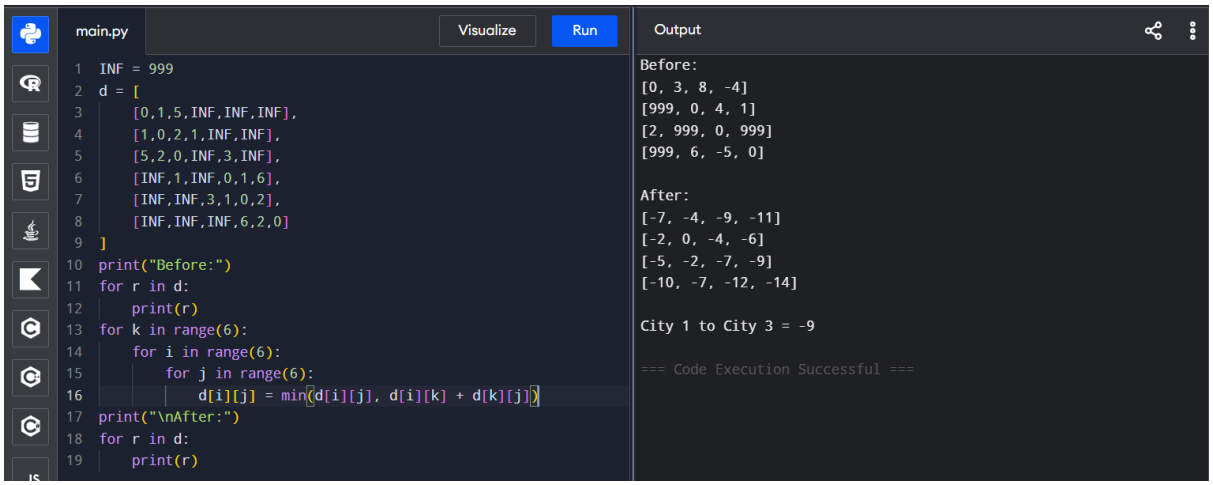
main.py

```python
class WordFilter:
    def __init__(self, words):
        self.words = words

    def f(self, pref, suff):
        for i in range(len(self.words)-1, -1, -1):
            if self.words[i].startswith(pref) and self.words
            [i].endswith(suff):
                return i
        return -1

w = WordFilter(["apple"])

print(w.f("a", "e"))
```

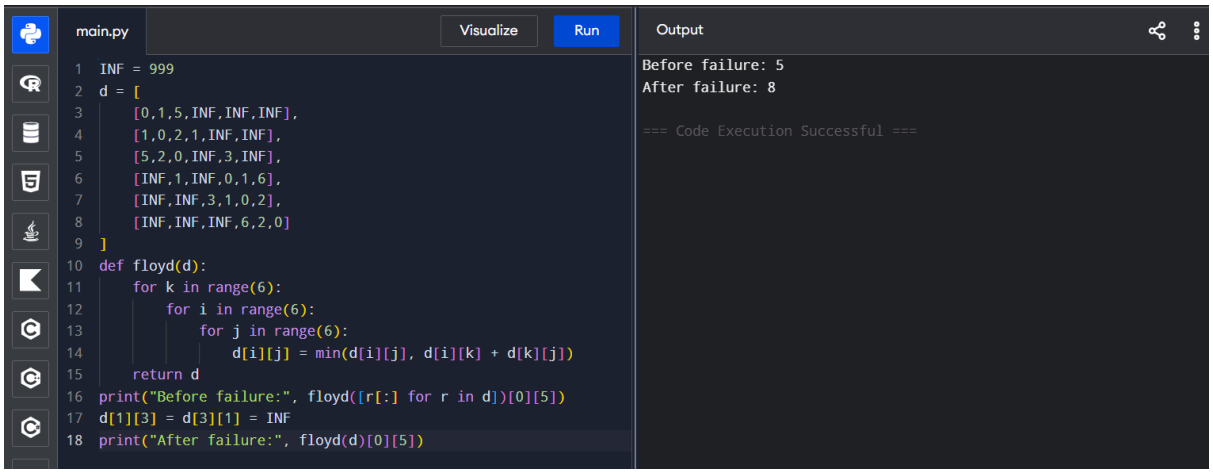
Output

```
0

=== Code Execution Successful ===
```

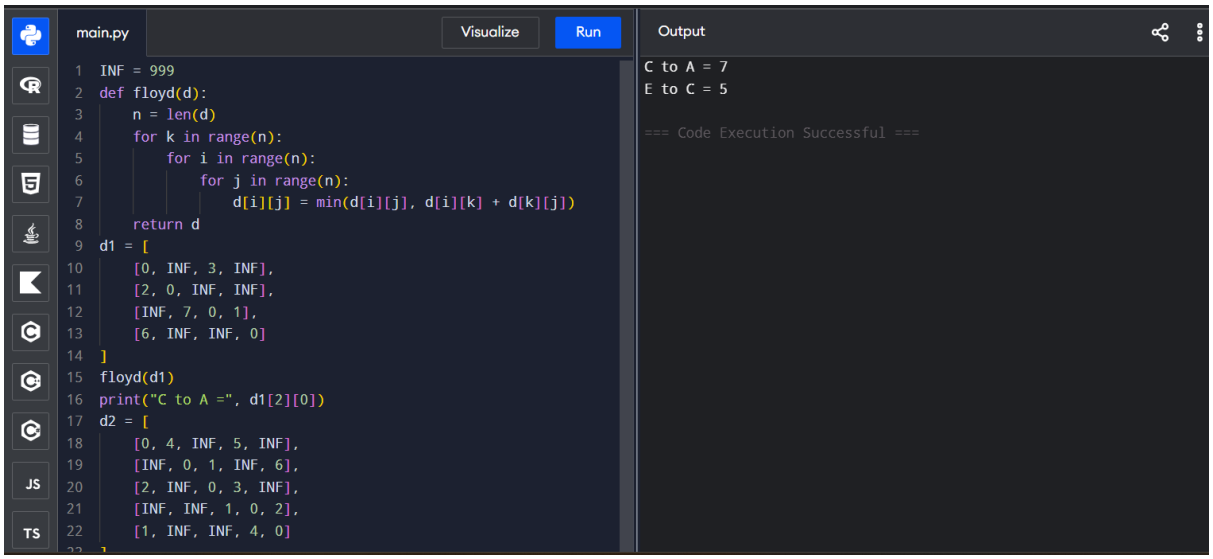
EXPERIMENT – 12



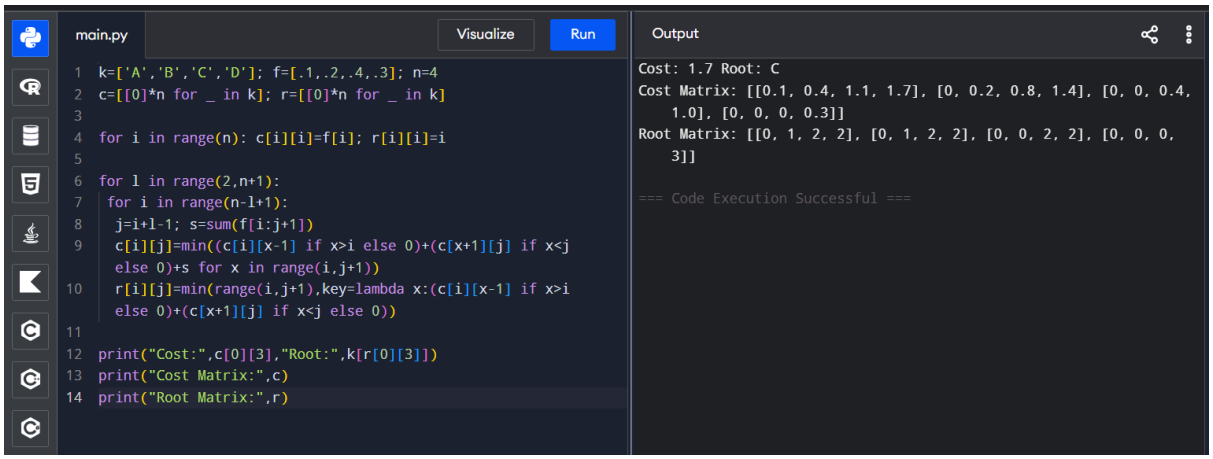
EXPERIMENT – 13



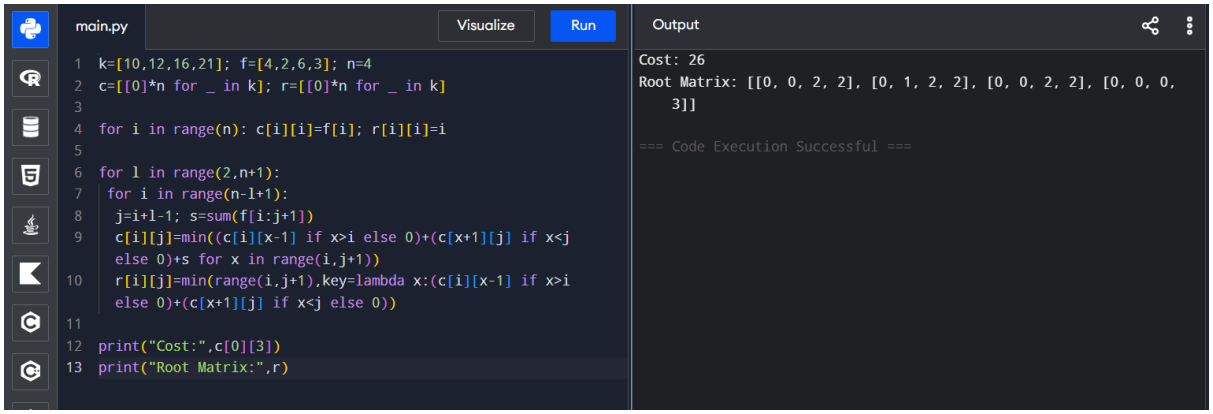
EXPERIMENT – 14



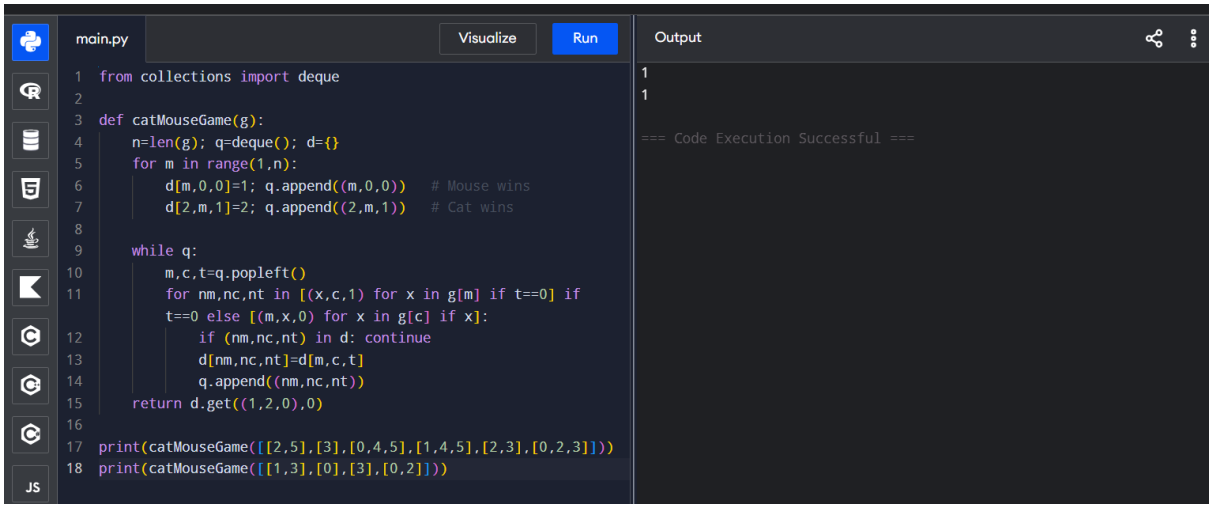
EXPERIMENT – 15



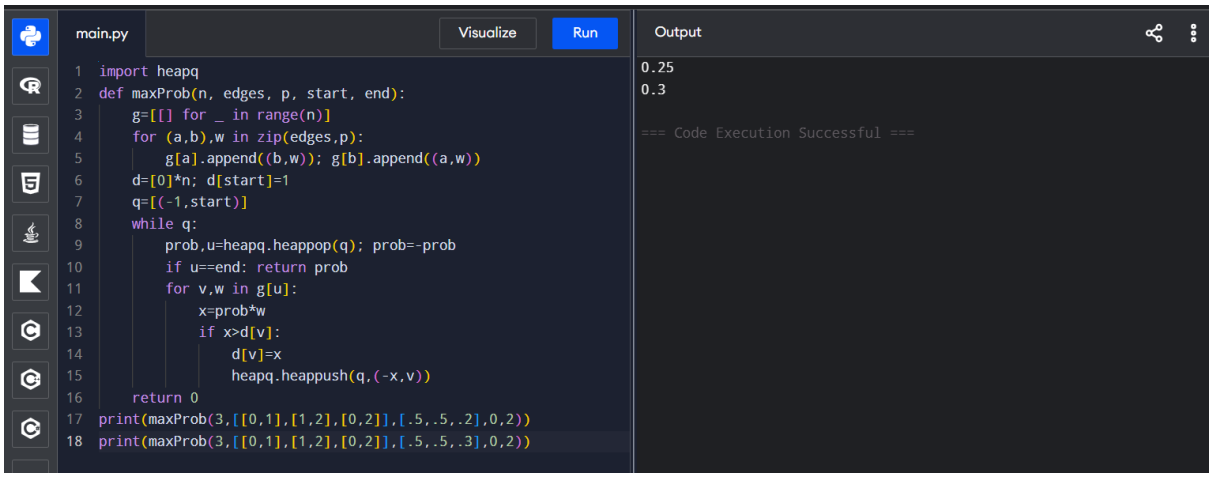
EXPERIMENT – 16



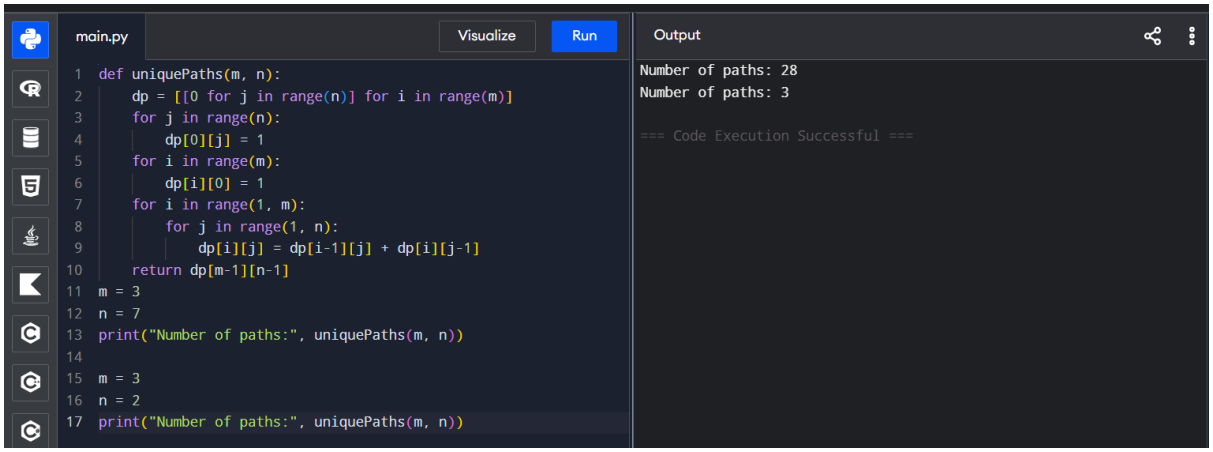
EXPERIMENT – 17



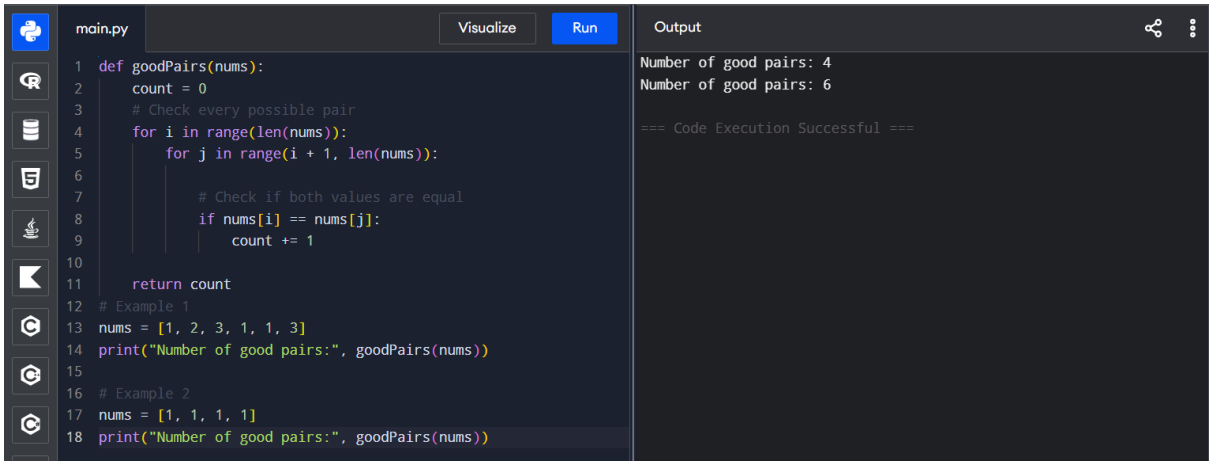
EXPERIMENT – 18



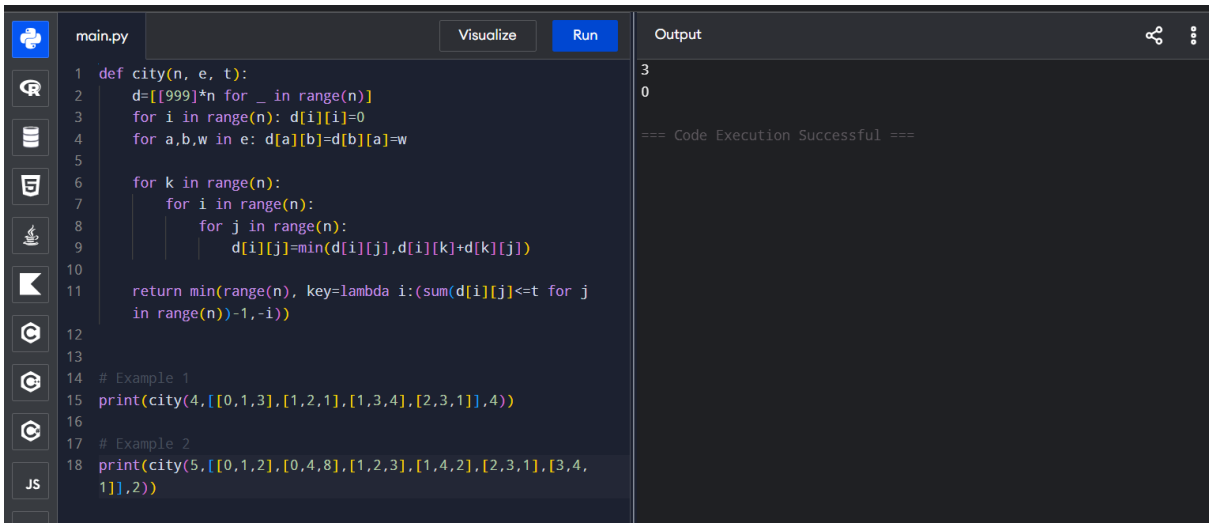
EXPERIMENT – 19



EXPERIMENT – 20



EXPERIMENT – 21



EXPERIMENT – 22